

Chapter 10 Warm-Up: Dictionaries and a Phone Book

Prof. Jordan C. Hanson

1 Dictionary Basics

1. A Python dictionary stores information as **key–value pairs**. Consider

```
phone_book = {  
    "562-555-0142": "Maria Garcia",  
    "562-555-0188": "Daniel Kim",  
    "562-555-0125": "Elena Garcia",  
    "562-555-0171": "Maria Lopez",  
    "562-555-0194": "Aisha Patel"  
}
```

Each phone number is a **key**. The corresponding name is its **value**. What is printed by

```
print(phone_book["562-555-0188"])
```

Why is it useful for each dictionary key to be unique?

2 Creating and Adding Entries

1. An empty dictionary can be created using

```
phone_book = {}
```

A new key–value pair can then be added:

```
phone_book["562-555-0142"] = "Maria Garcia"
```

Add another entry:

```
phone_book["562-555-0188"] = "Daniel Kim"
```

Print the entire dictionary:

```
print(phone_book)
```

What happens if you assign a new value to a key that already exists?

3 Dictionary Operations

1. The method `phone_book.keys()` returns the dictionary keys. The method `phone_book.values()` returns the values. The method `phone_book.items()` provides the key and value together. For our phone book (`phone`, `name`) represent one key–value pair.

4 Searching by Name

1. We can combine dictionaries with the string-searching techniques from Chapter 8. Begin with

```
search = input("Enter first/last name, or both: ")
```

Then examine each entry:

```
for phone, name in phone_book.items():  
    if search.lower() in name.lower():  
        print(name, phone)
```

Predict the output when the user enters

Garcia

How many phone numbers should be printed?

5 Put the Search in a Function

1. Write a function that returns all matching phone numbers:

```
def find_numbers(phone_book, search):  
    matches = []  
    ...  
    return matches
```

Test the function:

```
print(find_numbers(phone_book, "Maria"))
```

Which phone numbers should be returned?

6 Warm-Up Exercise

Using the original five-entry dictionary, predict the result of each search before running the program:

```
find_numbers(phone_book, "Garcia")  
find_numbers(phone_book, "Maria")  
find_numbers(phone_book, "Daniel Kim")  
find_numbers(phone_book, "Johnson")
```

Answer the following:

- Which search returns two phone numbers?
- Which search uses both a first and last name?
- What does the function return when there are no matches?
- Why is a list used for `matches` instead of returning the first phone number immediately?